# What a Function Cannot Tell Apart: Fibres, Collision Counts and the Inverse Problems of Finance

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**Abstract**

A non-injective function groups its inputs into sets that it cannot distinguish. Starting from the familiar practice of restricting the domain of $x \mapsto x^2$ in order to invert it, I study the structure induced on the inputs by their common image: fibres, the associated partition, the quotient set, and a simple collision count $I(f)$ with its behaviour under composition. These constructions are classical, and the link with the sheets of Riemann surfaces is explained. In finance, however, two inputs never have exactly the same image, so the relevant relation is equality up to a tolerance $\varepsilon$. It is not transitive, and it leads to a normalised count $p_\varepsilon$ and to ellipsoidal $\varepsilon$-fibres. For minimum-variance portfolios the $\varepsilon$-fibre is an exact ellipsoid, and a simulation study on 20 simulated assets shows that portfolios with almost identical variance can differ by more than 40% in turnover, with the disagreement concentrated in the flat eigen-directions of the covariance matrix.

**Keywords:** fibres, quotient set, indistinguishability, ill-posed inverse problems, portfolio optimisation, estimation error.

## 1 Introduction

The question that started this work came in a dull moment. During an internship in commerce and logistics, I returned to something that had always bothered me about inverse functions. Take $f(x) = x^2$ on $\mathbb{R}$. Since $f(2) = f(-2) = 4$, the function is not injective, and the usual way to obtain an inverse is to restrict the domain to $[0, +\infty)$, after which $f^{-1}(x) = \sqrt{x}$. But why should $-2$ be discarded because 2 has the same image? If $4 = f(2) = f(-2)$, the full inverse information would seem to be $4 \mapsto \{-2,2\}$. The question became more precise during a tram ride home: instead of asking how to force an inverse to exist, what structure does a function induce among the inputs it cannot distinguish?

Sections 2 to 4 develop this question in an abstract setting. They rebuild known objects (fibres, partitions, quotients) and add a collision count, studied under composition. Section 5 explains why the exact version cannot be used directly in finance and replaces equality by equality up to a tolerance. Section 6 lists the questions I wanted to answer, and Section 7 tests one of them on simulated portfolios. None of the abstract material is new to mathematics; I state clearly below which statements are standard.

## 2 Fibres, partitions and quotients

Let $f: X \to Y$ be a function. For $y \in Y$ the *fibre* of $y$ is $f^{-1}(y) = \{x \in X: f(x) = y\}$. Two inputs are indistinguishable when they lie in the same fibre.

**Definition 1**. For $x, x' \in X$ write $x \parallel_f x'$ if $x \neq x'$ and $f(x) = f(x')$.

I think of this as an inverted equality: the inputs remain distinct, but the function makes them equal at the level of their images. For $f(x) = x^2$ one has $2 \parallel_f -2$. The nonempty fibres form a partition $P_f = \{f^{-1}(y): y \in f(X)\}$ of $X$. The relation $x \sim_f x' \Leftrightarrow f(x) = f(x')$ is an equivalence relation whose classes are exactly the fibres, and $\parallel_f$ is its off-diagonal part, $\parallel_f = \sim_f \backslash \Delta_X$ with $\Delta_X = \{(x, x): x \in X\}$.

Forming the quotient set $X/\sim_f$ keeps each group of indistinguishable inputs as a single point instead of choosing one representative. Let $\pi_f: X \to X/\sim_f$ be the canonical projection $\pi_f(x) = [x]$ and define $\tilde{f}: X/\sim_f \to f(X)$ by $\tilde{f}([x]) = f(x)$.

**Proposition 1**. *The map $\tilde{f}$ is well defined and bijective, and $f = \tilde{f} \circ \pi_f$.*

This is the first isomorphism theorem for sets. A function fails to be injective exactly when some fibre has more than one element, that is, when several points of $X$ represent the same point of the quotient. For $f(x) = x^2$ on $\mathbb{R}$ the classes are $\{x, -x\}$ for $x \neq 0$ and $\{0\}$, and $\tilde{f}: \mathbb{R}/\sim_f \to [0, \infty)$ is a bijection, so the square root becomes a genuine inverse once one works on the quotient. Restricting the domain to $[0, \infty)$ does something different: it chooses one representative in each class.

*Remark 1*. For an arbitrary set $X$ the quotient $X/\sim_f$ is only a set. Speaking of a quotient *space* requires a topology, for instance the quotient topology in which $U$ is open when $\pi_f^{-1}(U)$ is open in $X$. At this point the subject leaves the elementary theory of functions and enters topology.

It is also useful to see the relation $\|_f$ as a graph. Take the elements of $X$ as vertices and join $x$ and $x'$ when $x \|_f x'$. Each fibre becomes a complete graph, so the graph is a disjoint union of complete graphs. For fibres of sizes 3,2,1 it is $K_3 \sqcup K_2 \sqcup K_1$.

## 3 A collision count

**Definition 2**. For finite $X$, let $I(f)$ be the number of unordered pairs of distinct inputs with the same image. If the fibres have sizes $m_1, \ldots, m_k$, then

$$I(f) = \sum_{i=1}^{k} \binom{m_i}{2} = \frac{1}{2}\left( \sum_{i=1}^{k} m_i^2 - |X| \right).$$

**Proposition 2**. *Let $f: X \to Y$ and $g: Y \to Z$ be maps with $X$ finite. Then $I(g \circ f) - I(f) = \sum_{\{y,y'\}} |f^{-1}(y)| \cdot |f^{-1}(y')|$, where the sum runs over unordered pairs of distinct points $y, y' \in f(X)$ with $g(y) = g(y')$. In particular $I(g \circ f) \geq I(f)$, with equality if and only if $g$ is injective on $f(X)$.*

*Proof.* A pair $\{x, x'\}$ of distinct inputs with $g\big(f(x)\big) = g\big(f(x')\big)$ is of one of two kinds. Either $f(x) = f(x')$, and the pair is counted in $I(f)$; or $y = f(x) \neq f(x') = y'$ and $g(y) = g(y')$. The pairs of the second kind with a given $\{y, y'\}$ are exactly the $|f^{-1}(y)| \cdot |f^{-1}(y')|$ choices of one point in each fibre. Summing over $\{y, y'\}$ gives the formula, and the right-hand side vanishes exactly when no two points of $f(X)$ are merged by $g$. □

Composition can merge fibres but it never splits them.

**Example 1**. Let $f$ have fibres of sizes 2,3,1, so that $I(f) = 1 + 3 + 0 = 4$. If $g$ merges the fibres of sizes 2 and 3, the new profile is 5,1 and $I(g \circ f) = 10$. The increase is $10 - 4 = 6 = 2 \cdot 3$.

For $r \geq 2$ one can define $I_r(f) = \sum_B \binom{|B|}{r}$, the number of indistinguishable $r$-tuples; since composition only coarsens the partition, $I_r(f) \leq I_r(g \circ f)$ for every $r$. For a chain $X = X_0 \to X_1 \to X_2 \to \cdots$ the partitions become coarser and the counts $I(F_k)$, with $F_k = f_k \circ \cdots \circ f_1$, are non-decreasing. With four elements, the chains $1 + 1 + 1 + 1 \to 2 + 1 + 1 \to 3 + 1 \to 4$ and $1 + 1 + 1 + 1 \to 2 + 1 + 1 \to 2 + 2 \to 4$ have counts 0,1,3,6 and 0,1,2,6 respectively; both end at $\binom{4}{2} = 6$, because each unordered pair becomes indistinguishable exactly once before total collapse.

*Remark 2*. The scalar $I(f)$ does not determine the partition, and it does not determine what can happen next. The profiles $4 + 1 + 1$ and $3 + 3$ both give $I = 6$, but from $4 + 1 + 1$ one can reach $5 + 1$ or

$4 + 2$, while from $3 + 3$ the only strict merger is 6. Likewise, on four elements $I = 2$ means $2 + 2$ and $I = 3$ means $3 + 1$, and $2 + 2$ has no coarsening with profile $3 + 1$. The state of the system is the partition, not the number.

## 4 Complex powers and Riemann's sheets

Over $\mathbb{C}$ the picture is richer. For $f(z) = z^n$ a non-zero value has $n$ preimages $z, \omega z, \dots, \omega^{n-1} z$ with $\omega = e^{2\pi i/n}$. Every non-zero fibre has size $n$ and corresponds to a $K_n$ in the graph; for $n = 2$ this is $z \leftrightarrow -z$.

This is where the relation with Riemann's treatment of multiple-valued functions becomes visible, a link I found only after writing the previous sections. The inverse $\sqrt{w}$ of $z \mapsto z^2$ is not a function: above each $w \neq 0$ there are two values $z$ and $-z$. Riemann's idea was to stop forcing a choice and to let the values live on sheets above the $w$-plane, joined at the branch point 0, so that the multivalued inverse becomes a single-valued function on a new surface. The correspondence with the present construction is direct. The multivalued inverse of $f$ is the relation $\{(y, x) : f(x) = y\}$, which as a set is $X$ itself, covering $f(X)$ through $f$. The sheets above $y$ are the elements of the fibre $f^{-1}(y)$, and $x \parallel_f x'$ says that $x$ and $x'$ lie on different sheets above the same point. Equality of images together with inequality of inputs is what I called an inverted equality. For $z^n$ there are $n$ sheets, and the branch point is the value at which the fibre shrinks, $z = 0$ for $z^2$, where the fibre has one element.

The difference is what each construction remembers. A Riemann surface carries a topology and a complex structure: the sheets are glued continuously. The fibres and the partition above are the set-level skeleton of that picture, with no gluing and no continuity, which is why a quotient *space* needs a topology (Remark 1).

## 5 Tolerance: indistinguishability in finance

In finance the map of interest sends the parameters of a model (a volatility model, say) to the prices it produces, or a portfolio to its estimated risk. Calibrating the model means going backwards from the quoted prices to the parameters. This is an inverse problem, and it is ill-posed in Hadamard's sense when the map is nearly non-injective: many parameter sets reproduce the quotes almost equally well, and a small perturbation of the quotes can move the solution a long way [1]. A regularisation rule [2] selects one element among them, which is the same gesture as restricting $x^2$ to $[0, \infty)$. It is a choice, and understanding the whole fibre shows how much of the answer comes from the data and how much from the rule.

The abstract setting does not transfer directly, because two parameter sets never give *exactly* equal prices; they give prices that differ by less than a spread or a numerical error. The honest relation is a tolerance relation.

**Definition 3**. For a metric $d$ on $Y$ and $\varepsilon > 0$ write $x \approx_\varepsilon x'$ if $d\big(f(x), f(x')\big) < \varepsilon$.

The relation $\approx_\varepsilon$ is reflexive and symmetric but *not transitive*. If three inputs have images 0, 0.6 and 1.2 and $\varepsilon = 0.7$, the first and second are indistinguishable, and so are the second and third, but the first and third are not. Hence there is no partition and no quotient, and the graph of indistinguishable pairs is no longer a disjoint union of complete graphs. This is the essential difference between exact and approximate indistinguishability.

The collision count survives in a modified form. For a finite set of candidates $X$ with $n$ elements, let $I_\varepsilon(f)$ be the number of unordered pairs with $d\big(f(x), f(x')\big) < \varepsilon$ and put

$$p_\varepsilon(f) = \frac{I_\varepsilon(f)}{\binom{n}{2}}.$$

It is non-decreasing in $\varepsilon$, and for $\varepsilon$ small enough $p_\varepsilon(f) = I(f)/\binom{n}{2}$. The map $\varepsilon \mapsto p_\varepsilon$ describes the candidate set at all resolutions, rather than by a single number.

*Remark 3.* If $q_i = m_i/n$ are the relative fibre sizes, the probability that two independent uniform inputs have the same image is $\sum_i q_i^2 = (2I(f) + n)/n^2$. This is the collision probability of the output, closely related to Simpson's concentration index [3] and to Rényi's entropy of order two [4]; for portfolio weights $w_i$ the quantity $1/\sum_i w_i^2$ is the usual effective number of assets. I did not know this when defining $I(f)$, and it shows that the count is a standard concentration measure in another guise.

**The geometry for portfolios**

Let $\Sigma$ be a positive definite covariance matrix and let $w^*$ minimise $V(w) = w^\top \Sigma w$ under $\mathbf{1}^\top w = 1$, with $v^* = V(w^*)$.

**Proposition 3**. *For every $w$ with $\mathbf{1}^\top w = 1$, $V(w) - V(w^*) = (w - w^*)^\top \Sigma\, (w - w^*)$. Consequently $\{w\colon \mathbf{1}^\top w = 1,\ V(w) \le (1+\varepsilon)v^*\} = \{w^* + \Delta\colon\ \mathbf{1}^\top \Delta = 0,\ \Delta^\top \Sigma \Delta \le \varepsilon v^*\}$ is an ellipsoid in the hyperplane $\mathbf{1}^\top \Delta = 0$.*

*Proof.* Write $w = w^* + \Delta$ with $\mathbf{1}^\top \Delta = 0$. Then $V(w) = v^* + 2\Delta^\top \Sigma w^* + \Delta^\top \Sigma \Delta$. The first-order condition for $w^*$ is $\Sigma w^* = \lambda \mathbf{1}$ for a multiplier $\lambda$, hence $\Delta^\top \Sigma w^* = \lambda \mathbf{1}^\top \Delta = 0$. □

Directions in which $\Sigma$ has small eigenvalues are therefore long axes of the ellipsoid: a portfolio can move far in those directions without the objective noticing. This is the portfolio version of the fibre.

## 6 Questions

The framework suggests three questions.

1. Can the collision count $I(f)$, or its tolerance version $I_\varepsilon(f)$, measure how many parameter sets or portfolios a calibration or an optimisation cannot tell apart, on a finite set of candidates?

2. In calibration problems, how large and how structured is the set of parameters that give the same prices, and how does a regularisation rule select a point in it?

3. When the inputs carry geometric structure, what space remains after the quotient $X/\!\sim_f$, and how does it relate to the sheets of Section 4?

I address Q1 for portfolios. Q2 requires a real option-pricing model and market data, and Q3 requires topology that I have not studied.

## 7 Experiment: portfolios the data cannot tell apart

### 7.1 Set-up

Here $f$ maps a portfolio $w$ to its in-sample variance $w^\top \Sigma w$, with $\Sigma$ estimated from the data, and $X$ is a finite set of candidates. I simulated $T = 750$ daily returns for 20 assets driven by one strong and one weaker common factor plus idiosyncratic noise, and computed the unconstrained minimum-variance portfolio (short selling allowed). A candidate set of 300 portfolios was obtained by recomputing the minimum-variance portfolio on 300 bootstrap resamples of the data. Each candidate is evaluated with the *original* covariance matrix, which is the one the optimiser saw. Exact collisions do not occur, so two candidates collide when their variances differ by less than $\varepsilon$ in relative terms. The distance between

two portfolios is measured by turnover, the fraction of the portfolio that must be traded to move from one to the other. All data are simulated; the code is given in the appendix.

### 7.2 Results

The sample covariance matrix has condition number 46.6: one dominant eigenvalue ($2.2 \times 10^{-3}$) and nineteen others between $4.7 \times 10^{-5}$ and $1.5 \times 10^{-4}$. No candidate has variance within 1% of the minimum, but 98% are within 5% and all within 10%. Table 1 gives the collision curve. At $\varepsilon = 2\%$, 87% of pairs are indistinguishable, yet their median turnover is 43%.

Table 1. Collision curve for 300 candidate portfolios ($T = 750$, 20 assets).

| $\varepsilon$ (relative variance) | $p_\varepsilon$ | median turnover of colliding pairs |
|---|---|---|
| 0.5% | 0.30 | 42% |
| 1% | 0.56 | 43% |
| 2% | 0.87 | 43% |
| 5% | 1.00 | 44% |

Decomposing the differences $w - w^*$ along the eigenvectors of $\Sigma$ shows that 97% of the squared deviation lies in the 18 flat directions and only 3% in the two high-variance directions (Figure 1), as Proposition 3 suggests. (The eigenvectors of $\Sigma$ itself, rather than of $\Sigma$ restricted to the hyperplane $\mathbf{1}^\top \Delta = 0$, are used here as an approximation.)

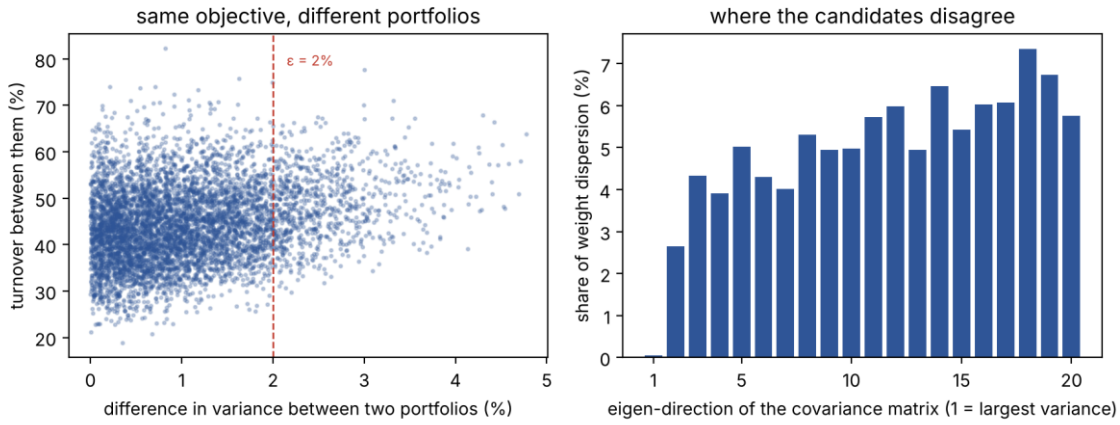



Figure 1. Left: each point is a pair of candidate portfolios (difference in variance against turnover). Right: share of the weight dispersion along each eigen-direction of the covariance matrix.

### 7.3 Dependence on the sample length

The experiment was repeated with 250, 750 and 2500 days, each with a fresh simulation (so the middle row differs slightly from Table 1). With more data the candidates converge, but the flat directions do not disappear.

Table 2. Collision probability and turnover for different sample lengths.

| $T$ | $p_{2\%}$ | median turnover of colliding pairs | mean turnover to $w^*$ |
|---|---|---|---|
| 250 | 0.35 | 74% | 54% |
| 750 | 0.88 | 44% | 32% |
| 2500 | 1.00 | 24% | 17% |

### 7.4 Out-of-sample cost

On 5000 fresh simulated days the sample minimum-variance portfolio has variance $8.22 \times 10^{-5}$, the average bootstrap candidate $8.42 \times 10^{-5}$ (2.5% higher), and the equal-weight portfolio $1.10 \times 10^{-4}$. Shrinking the covariance matrix towards a diagonal matrix did not improve on the sample minimum-variance portfolio ($8.42 \times 10^{-5}$ with intensity 0.2 and $9.34 \times 10^{-5}$ with 0.5) [6].

## 8 Discussion

The normalised tolerance count $p_\varepsilon$ does measure how many candidate portfolios the objective cannot tell apart, and the collision curve is more informative than a single number. In the experiment it reveals something the objective alone hides: portfolios that the data cannot separate are far from each other in weights, and the distance lies almost entirely in the flat eigen-directions, as the ellipsoid picture predicts. The fragility is thus real in the weights, and so in turnover and transaction costs, but it is cheap in risk: the candidates cost only about 2.5% extra variance out of sample. This qualifies the common statement that optimised weights are fragile, which is true of the weights and much less true of the risk [5]. I also expected shrinkage to help, and it did not; with 750 observations for 20 assets the estimation error is modest, and I did not test the regime with many more assets than observations.

The limits are clear. The data are simulated, and the factor structure and the tolerances are my choices, so the qualitative picture should carry over but not the numbers. The candidates come from a bootstrap and are not a uniform sample of the $\varepsilon$-fibre, so $p_\varepsilon$ describes the candidate set as much as the data; the choice of $\varepsilon$ is arbitrary, which is why the whole curve matters. Finally, none of this is new as mathematics or as finance: the same phenomena are studied under the names of identifiability, ill-posedness and resampled efficiency. What I have done is to arrive at them from a different starting point and to measure them with a collision count. The natural next steps are to repeat the analysis on real return data, with long-only constraints and transaction costs, to examine the regime of many assets and few observations, and to turn to Q2 with a real option-pricing model.

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## Appendix: Reproducing the experiment

All numbers and the figure were produced by a Python script . The simulated returns can be replaced by a $T \times 20$ array of real daily returns to run the same analysis on actual data.